Verifold

A zero knowledge proof verifier for Stacks, written entirely in Clarity.
Build and test metrics, verifiable from the public repository.
Snapshot date: 16 June 2026 · **working prototype, demonstration parameters**

Summary

A complete zero knowledge proof, produced by an independent prover, is verified end to end by the Clarity contract on a simulated network running Stacks' real cost model, at demonstration settings, using about one percent of a single block. The work below is reproducible by cloning the repository and running the test suite and the cross check harness.

The verifier

Metric	Value
Clarity contracts	**14**
Total Clarity lines	**2,157**
Verifier logic (13 contracts, excluding the generated cost artifact)	**~1,468 lines**
Generated cost measurement artifact (fullsize.clar)	**689 lines**
Consensus change required	**None**

Built on primitives Clarity already exposes: sha256 hashing and native 128 bit integer arithmetic. No SIP, no fork, no protocol upgrade.

Automated tests

Metric	Value
Total automated tests	**275**
Passing	**275 (100%)**
Test files	**16**
Grouped test sections	**89**

Two of the 275 are compute heavy property tests that exceed the default five second per test limit on slower machines; both pass when the timeout is raised. Tests are differential against independent TypeScript reference implementations, pinned by known answer vectors, and include a large adversarial negative matrix where every invalid proof must be rejected.

Independent cross checks

- **Rust cross check harness against StarkWare's Stwo library** (about 530 lines of Rust). Validates the verifier's conventions across five labeled groups (domain, interpolation and openings, quotients, DEEP, and FRI folding), value for value against Stwo's own functions.

- **Independent Rust mini prover** (327 lines) that produces real proofs the Clarity contract accepts; proof fixtures stored in the repository.
- **Independent Python replay** that pins the known answer vectors from scratch, plus the full parameter cost shape generator.
- **Three independent implementations** (Clarity, TypeScript, Python) plus the Rust prover and the Stwo cross check, so a bug must appear identically in several places to slip through.

Measured cost (Stacks real cost model, via Clarinet)

Metric	Value
Per block runtime budget	**5,000,000,000 units**
One verify at demonstration settings	**~1.0% of a block**
Full strength verify (cost artifact, across realistic settings)	**4% to 10% of a block**

Cost figures are point in time under the current Stacks cost model. The full strength, single transaction artifact is the next milestone and does not yet exist; the figure above is a measured cost shape, not a deployed verifier.

Status

Working prototype at demonstration parameters. Not audited and not production secure. No security claim is made before raised parameters, independent expert review, and a professional audit. The honest detail is in the project litepapers.

All figures above are reproducible from the public Verifold repository. Test counts are from a full run of the suite; contract and line counts are direct measurements of the source tree; cost figures are produced by Clarinet against Stacks' real on chain cost model.